

● FYDP-I · PROPOSAL DEFENCE

Depot

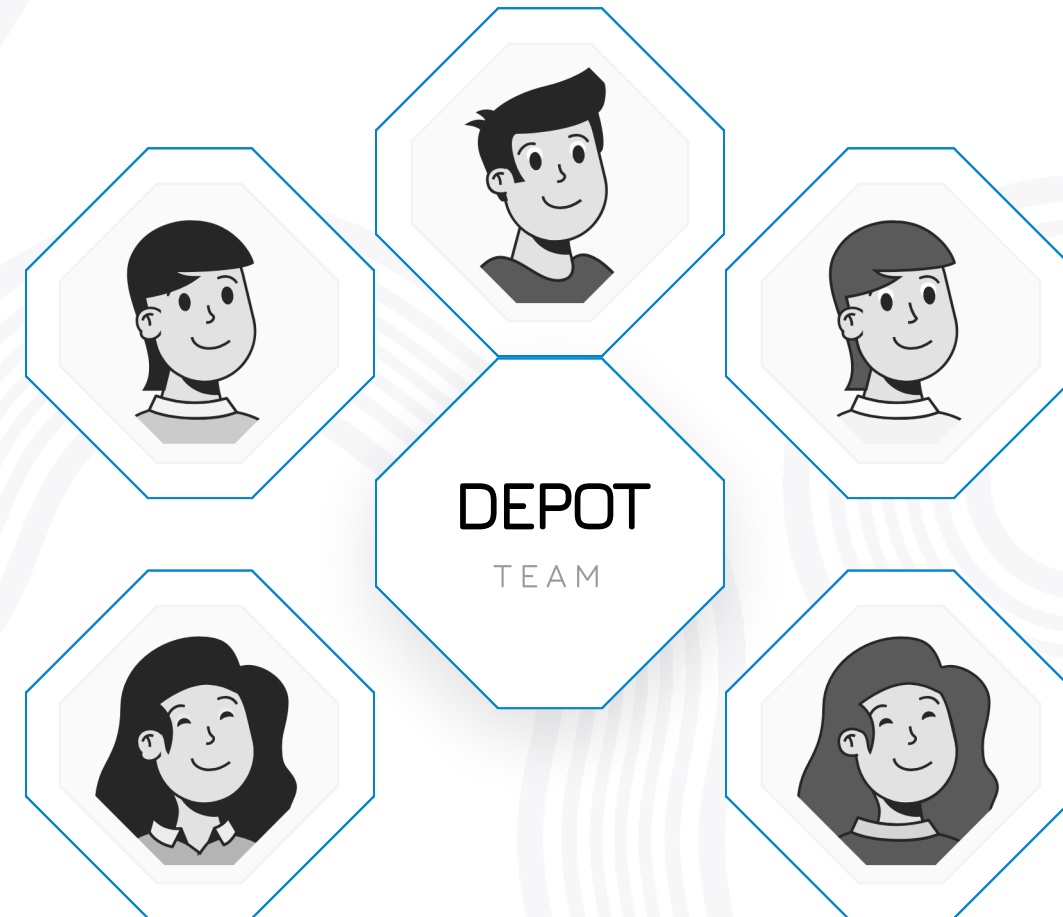
An AI-powered infrastructure architect,
from code to cloud.

TEAM

Ahsan Riaz
Muhammad Waleed
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ADVISORS

Hira Anwar
Ayesha Hakim



• THE CHALLENGE

Building is fast now.
Running it is still expert work.

~95%

of the code was AI-generated in a quarter of YC's Winter 2025 start-ups [1]

150%

average yearly growth in infrastructure-as-code questions on Stack Overflow, 2011–2022 [2]

0%

of one model's Terraform passed a security scan, though 77.8% passed validation [3]

● PROBLEM STATEMENT

Working code still leaves four decisions.

01

What it needs

02

Where it runs

03

What it costs

04

How to change it
safely

Today's tools answer one or two. None takes an application as it is, weighs clouds against managed platforms, and changes nothing without the user's approval.

Depot answers all four, **and explains, records and approves every change.**

• WHY IT MATTERS

Many build on the cloud. Few can run it safely.

CLOUD USE VS. IAC USE

43.3%

worked extensively with AWS

17.8%

with Terraform

Stack Overflow Developer Survey 2025 [4]

GPT-4 PASS RATE

86.6%

on general coding tasks

19.36%

on infrastructure code

IaC-Eval, NeurIPS 2024 [5]

CLOUD SPEND

84%

call managing it their top cloud challenge

29%

of IaaS and PaaS spend estimated wasted

Flexera, 2025 and 2026 [6] [7]

• WHAT EXISTS TODAY

Each tool solves part of it.
Someone still decides where it runs.

	Starts from code as it is	Runs in the user's account	Chooses cloud or platform	Approval before change
Managed platforms Vercel, Railway, Render	✓	—	—	½
Bring-your-own-cloud Porter, Qovery, Northflank, Ravion	½	✓	—	½
Infrastructure-from-code Encore, Nitric, SST, Defang	—	✓	—	—
AI for one provider or codebase Pulumi Neo, StackGen, Gemini Cloud Assist, azd init	½	½	½	½
Depot	✓	✓	✓	✓

Fifteen products, from their own documentation, 30 Sep 2026 [8]. ½ = partly, or only some products; — = no, or not stated. Depot is planned.

● OBJECTIVES

Eight objectives. Two of them are research.

01 **Model** the application from its code and a short intake

03 **Deploy** to five targets with scoped credentials

05 **Operate** keep it running, inspectable and affordable

07 **Compare** language models on extraction, selection and explanation **RESEARCH**

02 **Select** the target and pattern that fit at the lowest cost

04 **Protect** against any unapproved or policy-violating change

06 **Agent** conversational changes and incident loop, with measured guardrails **RESEARCH**

08 **Validate** requirements and usability with target users

- USERS OF THE SYSTEM

Built for builders without an infrastructure engineer.



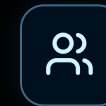
AI-assisted builders

A working product, but little experience of running one.



Solo full-stack developers

Outgrowing a managed platform: stay, or move to a cloud?

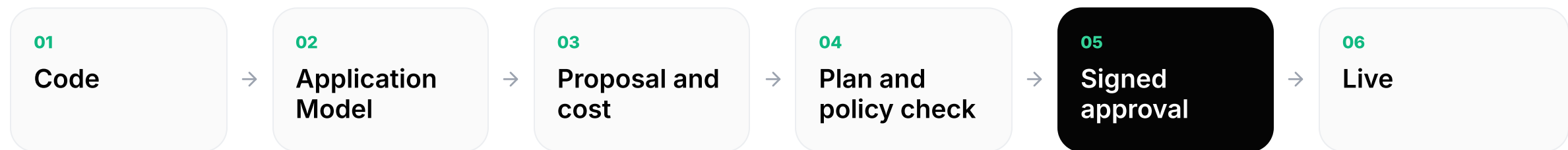


Small teams and agencies

Two to ten people, with no DevOps staff.

● HOW DEPOT WORKS

From code to live, through one gate.



Every later change returns through the same gate

CONFIRMATION MATCHES THE RISK

Read logs or metrics
runs at once

Restart or scale within limits
one confirmation

Change infrastructure
signed approval

The model selects; reviewed templates execute. The checks, not the generation, do the work. [9]

 KEY FEATURES

What Depot actually does.



Application Model

Every fact marked detected, inferred or user-supplied.



No stored keys

Bounded roles, federation, or one scoped token per target.



Architect

Picks among reviewed patterns, ranked by cost and effort.



Delivery and operations

Deploy on push, rollback, logs, cost, alerts, savings.



Safety gate

Validate, saved plan, policy, cost and role check.



Agent and incident loop

Changes by conversation; alarms end in a fix a person approves.

● DEPOT IN ACTION

One session, from working code to live.

STEP 1

Connect

GitHub, GitLab or an archive

STEP 2

Answer

Up to eight questions

STEP 3

Review

Target, architecture, monthly cost

STEP 4

Approve

Signed and role-checked

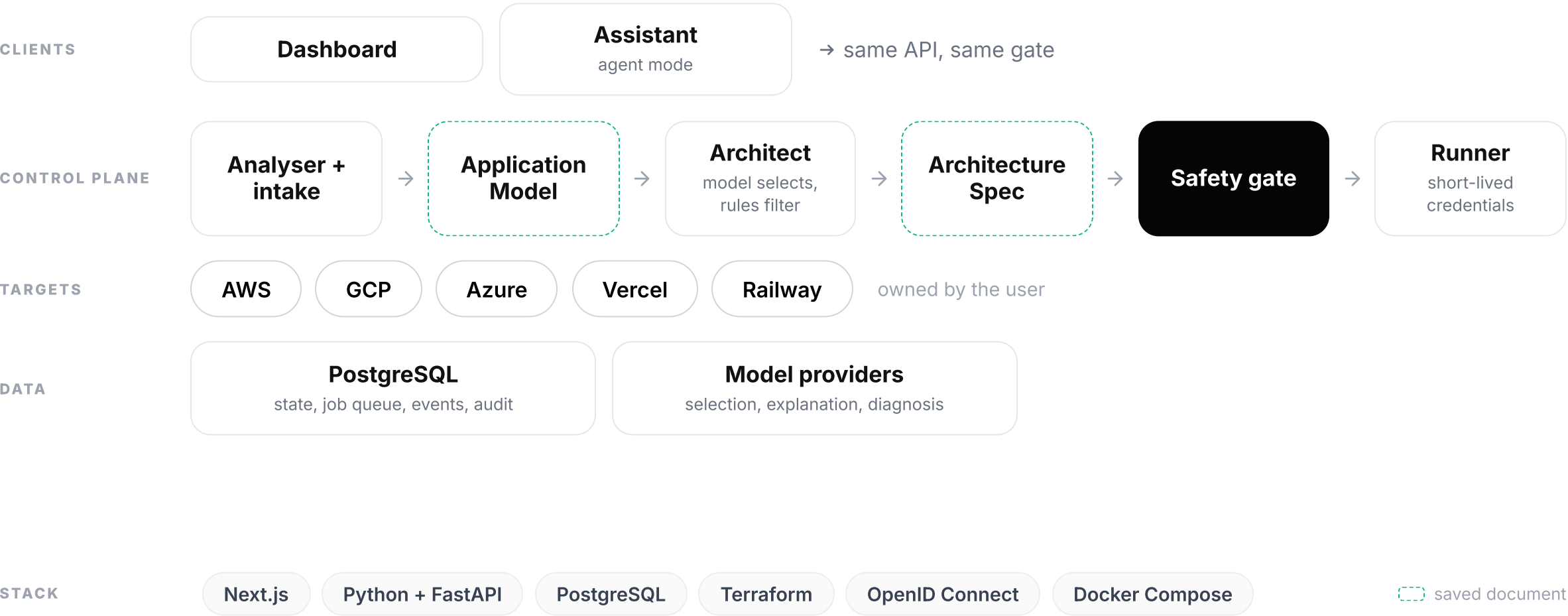
STEP 5

Run

Deploy on push, roll back, watch cost

SYSTEM ARCHITECTURE

One API, one gate, two saved documents.



● SCOPE

In scope, and deliberately out.

IN SCOPE

- ✓ GitHub, GitLab or an archive
- ✓ Web apps, APIs, static sites and workers (JS/TS, Python, Go)
- ✓ AWS, GCP, Azure, Vercel, Railway
- ✓ Scoped, short-lived credentials
- ✓ Approval before every infrastructure change
- ✓ Delivery, rollback, monitoring, cost, savings, agent

OUT OF SCOPE

- ✗ GPU, mobile and desktop software
- ✗ Clouds without a reviewed pattern library
- ✗ Pasted long-lived keys
- ✗ Any unapproved change
- ✗ Compliance certification

Extended scope only if M8 is met early: guided mode, Oracle and Yandex Cloud, voice, preview environments.

● EVALUATION

How we'll prove it works.

01 Model fields match the reference $\geq 85\%$

03 Apps healthy after one approved apply ≥ 7 of 8

05 Push-to-live and rollback median ≤ 10 min

08 Usability, System Usability Scale [10] ≥ 68 [11]

02 Right target and pattern on real apps $\geq 80\%$

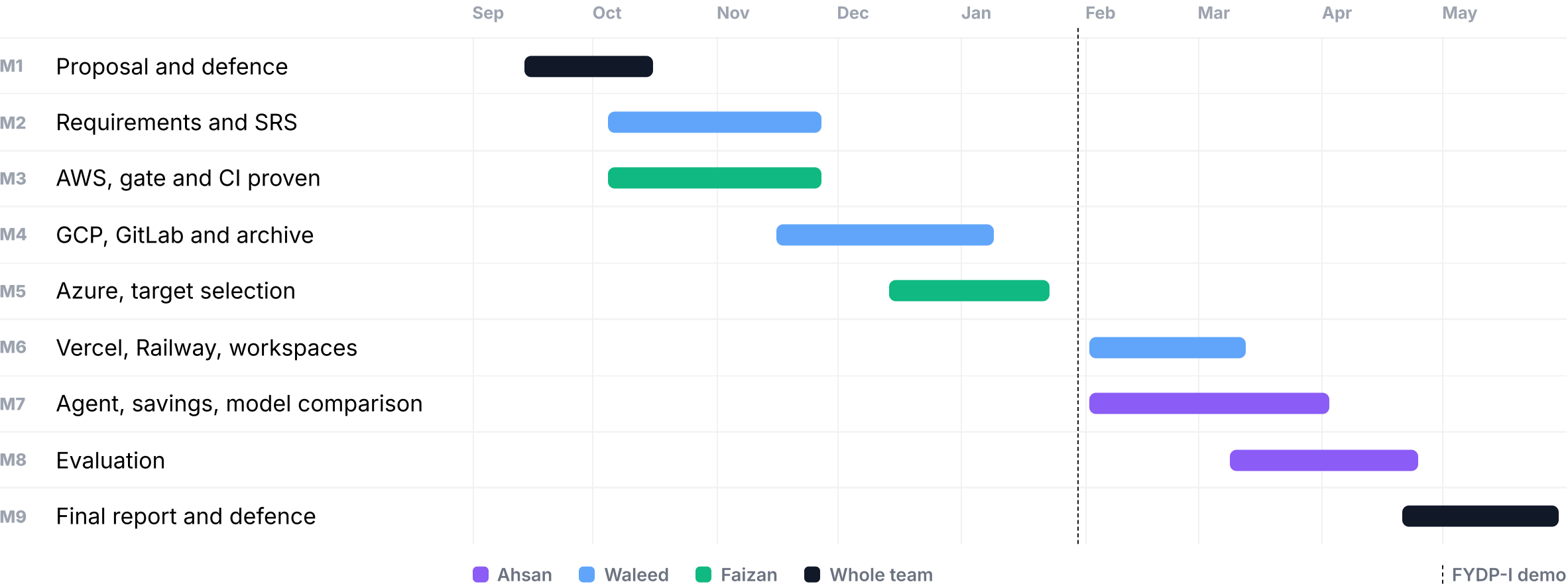
04 Changes run without a valid approval 0 HARD

06 Adversarial cases that get through ($n \geq 100$) 0 HARD

Tested on ≥ 30 independently labelled public apps, 8 deployments with injected faults, 5 baselines and ≥ 8 target users. Missing a hard threshold fails the objective.

TIMELINE

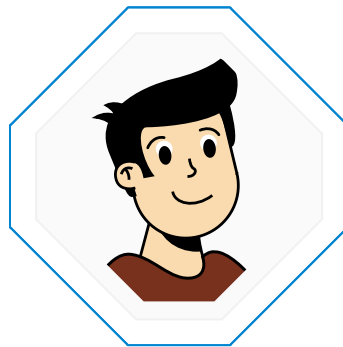
Three clouds first, then everything else.



Team planning targets; dates will follow the official defence schedule.

The Architects

Three owners. No member reviews only their own work.



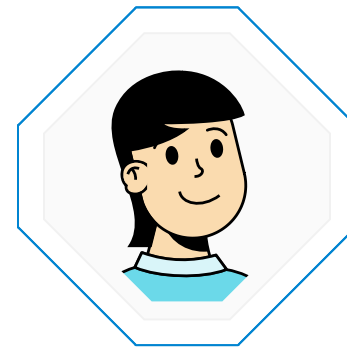
Ahsan Riaz

Application Model, agent mode,
incident loop, research

01

06

07

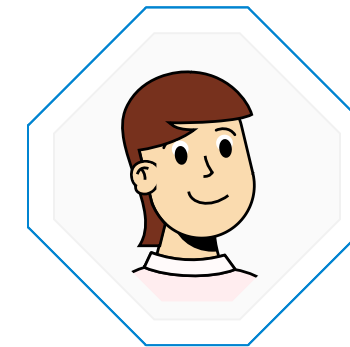


Muhammad Waleed

User-facing work, team workspaces,
managed platforms

05

08



Muhammad Faizan Anwar

Architect, cloud adapters, safety gate,
savings

02

03

04

Advisor **Hira Anwar** · Co-advisor **Ayesha Hakim** · Scrum, two-week sprints

• RISKS AND SAFETY

What could go wrong, and how we handle it.

Cloud accounts or credits not ready	First-sprint task with an owner; no features until fixed
Scope outgrows three people	Extended scope only if M8 is met early; Railway can move out
AI proposes unsafe infrastructure	The model only selects; reviewed templates execute
An agent is tricked into acting	Fixed actions, fixed permissions, approval by risk [12]
Credentials reach too far	Bounded, short-lived roles from M3; no stored user keys
A member acts without the role	Server-side role check on every route

Every change is traceable to its model, plan and approval.

● UN SUSTAINABLE DEVELOPMENT GOALS

Two goals we contribute to.

8 DECENT WORK AND ECONOMIC GROWTH



SDG 8.3

Small-enterprise growth

Small teams run products on cloud accounts they own, at a known monthly cost, without hiring.

9 INDUSTRY, INNOVATION AND INFRASTRUCTURE



SDG 9.5

Research and technological capability

An open benchmark, and views that explain why each architecture was chosen.

● REFERENCES

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• DEPOT · FYDP-I

Thank you.

Any questions?

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